

Problem 8.11

A fund earns interest at a rate equivalent to the rate of discount of d . Megan invested \$10,000 in the fund. Eleven years later, Megan has \$30,042. Calculate d .

Answer

The accumulated value equation is

$$10,000 \left(\frac{1}{1-d} \right)^{11} = 30,042.$$

Dividing by 10,000,

$$\frac{1}{(1-d)^{11}} = 3.0042.$$

Thus,

$$\frac{1}{3.0042} = (1-d)^{11}.$$

Taking the eleventh root,

$$\left(\frac{1}{3.0042} \right)^{1/11} = 1-d.$$

Therefore,

$$d = 1 - \left(\frac{1}{3.0042} \right)^{1/11} = 0.09516351.$$

Hence,

$$d = 9.516\%.$$